

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Sondasi
Program/Batch	B083 - Networkwalks
Date	17 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems and devices where I had secured written permission or the devices/systems that I own myself. All of these materials are for education and research purposes only. Do not use anything from this report to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action taken is my own responsibility. Misuse can lead to criminal charges, heavy fines, loss of employment and a permanent record. In most countries, unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. This is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning activities

WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate DNS records (SOA, NS, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the **networkwalks.com** domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** (whois networkwalks.com) to obtain publicly available domain registration information. The results showed the domain is registered with GoDaddy.com, LLC, created on 6 November 2019 and set to expire on 6 November 2027, with clientDeleteProhibited/RenewProhibited/TransferProhibited/UpdateProhibited locks in place and DNSSEC unsigned. The domain uses HostGator name servers (NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM), which identifies the actual hosting infrastructure behind the site.

I then used **WhatWeb** (whatweb networkwalks.com) to identify technologies used by the website. The results identified WordPress 7.1 and WP Download Manager 3.3.58, along with the server IP address 192.232.216.135, the Apache server, Bootstrap 7.1, jQuery 3.7.1, an exposed contact email (info@networkwalks.com), and a set of uncommon HTTP headers. The plain HTTP request also showed a 301 redirect to the HTTPS version of the site.

Using **Nslookup** I attempted to resolve the domain name to its IP address. The command was run against a mistyped hostname (networkswalks.com) and returned NXDOMAIN, confirming that DNS correctly returns no record for a name that does not exist. The correct IP address for networkwalks.com, 192.232.216.135, was independently confirmed through the WhatWeb and Curl results below. This is noted here as a methodology lesson: a single-character typo in a domain name is enough to produce a misleading “no record” result during reconnaissance.

I used **Curl** with the -I option (curl -I https://networkwalks.com) to inspect the HTTP response headers. The site returned HTTP/2 200 from an Apache server, set a __wpdm_client cookie, and exposed the WordPress REST API link (/wp-json/, including a specific /wp-json/wp/v2/pages/53 endpoint). The response also included a Permissions-Policy header referencing Google, Cloudflare and hCaptcha domains, and cache headers (x-endurance-cache-level, x-nginx-cache: WordPress) that reveal the hosting/caching stack.

Next, I used **Wafw00f** (wafw00f networkwalks.com) to determine whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs) as the active WAF, detected after 2 requests.

Finally, I used **DNSRecon** (dnsrecon -d networkwalk.com) to enumerate DNS records. As run, the command targeted a mistyped domain (networkwalk.com, missing the final “s”). The enumeration nonetheless completed and returned SOA and NS records pointing to ns35/ns36.domaincontrol.com (GoDaddy name servers) rather than the HostGator name servers seen in the WHOIS record for the real networkwalks.com domain — no SRV records were found. This mismatch confirms the DNSRecon query resolved a different, similarly-named domain rather

than the intended target, reinforcing the same lesson observed with Nslookup: exact domain spelling matters during reconnaissance, and results should always be cross-checked against a second source (in this case, WHOIS) before being relied upon.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap to perform network discovery on my local network. The practical required identifying the local subnet, discovering live hosts, identifying their IP and MAC addresses, and generating a network topology.

I entered the subnet 10.0.3.2/24 into Zenmap and selected the **Ping scan** profile, which ran the command `nmap -sn 10.0.3.2/24`. The scan completed in 11.64 seconds and identified 3 live hosts out of the 256 addresses in the subnet:

- 10.0.3.2 — up, MAC address 52:54:00:12:35:00 (QEMU virtual NIC)
- 10.0.3.3 — up, MAC address 52:54:00:12:35:00 (QEMU virtual NIC)
- 10.0.3.15 — up (localhost / scanning host, no MAC reported for the local interface)

After completing the scan, I opened the Topology section in Zenmap, which displayed the three live hosts connected to the localhost node on a fisheye ring diagram, confirming the same three addresses found in the Nmap output.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology & version exposed	WhatWeb identified WordPress 7.1 and WP Download Manager 3.3.58 on networkwalks.com	Attackers may target exposed CMS/plugin versions for known vulnerabilities	● Medium
2	Server IP address identifiable	WhatWeb and curl both resolved the site to 192.232.216.135	Reveals hosting location/provider, aiding further enumeration	● Low
3	HTTP technical information exposed	curl -I returned server, cookie and header details and exposed the /wp-json/ REST API endpoint	May assist technology fingerprinting and API-level enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs) protecting the site	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon enumeration returned SOA and NS records for the queried domain	DNS information can help build a broader infrastructure profile	● Medium
6	Domain registration details exposed	WHOIS returned registrar (GoDaddy), creation/expiry dates and HostGator name servers for networkwalks.com	Supports social-engineering and infrastructure-mapping efforts	● Low
7	Multiple live hosts visible on local network	Zenmap ping scan of 10.0.3.2/24 identified 3 live hosts (10.0.3.2, 10.0.3.3, 10.0.3.15) with MAC addresses	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorised security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information** — Networkwalks should periodically review what information about its web technologies, CMS and plugins (WordPress 7.1, WP Download Manager 3.3.58) is publicly visible via tools such as WhatWeb.
2. **Keep software updated** — WordPress core and installed plugins should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers** — The exposed /wp-json/ REST endpoint and caching headers should be reviewed to confirm no unnecessary technical detail is being disclosed.
4. **Review DNS and WHOIS records regularly** — Name-server and registration details should be checked periodically to ensure only required information is publicly exposed.
5. **Properly configure and monitor the WAF** — Keep ModSecurity (SpiderLabs) enabled and tuned, since it already blocks naive automated attacks.
6. **Perform regular internal network discovery** — Periodically scan the local network to keep an up-to-date inventory of active devices.
7. **Investigate unknown devices** — Any unexpected device discovered during a network scan should be investigated and verified.
8. **Double-check target spelling during reconnaissance** — As seen with the Nslookup and DNSRecon steps in this exercise, a single mistyped character can silently redirect a query to the wrong domain; always verify the target name and cross-check results against a second source.
9. **Maintain network documentation** — Network topology and device information should be documented and updated regularly.
10. **Perform security testing with authorisation** — Reconnaissance and scanning should only be performed against systems and networks where appropriate authorisation has been granted.

7. Conclusion

During Week 2 of my Cybersecurity internship (Batch B083), I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the networkwalks.com domain. I learned how WHOIS can provide domain registration and name-server information, WhatWeb can identify web technologies and exposed CMS/plugin versions, Nslookup can resolve (or fail to resolve) a domain name, Curl can inspect HTTP headers and exposed API endpoints, Wafw00f can identify a WAF, and DNSRecon can enumerate DNS records. I also learned first-hand, through the mistyped Nslookup and DNSRecon queries, how easily a small typo can produce a misleading result during reconnaissance, and why cross-checking findings against a second tool matters.

In the network scanning activity, I used Zenmap to scan my local subnet (10.0.3.2/24) and identify three active hosts, together with their IP and MAC addresses, and generated a network topology diagram confirming the discovered hosts.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analysing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorised scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

Evidence 1 — WHOIS lookup (whois networkwalks.com)

```
kali@kali:~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 245219295-DOMAIN_COM-VERIS
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2022-11-12T18:08:42Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2495
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint form: https://www.icann.org/wicf/
>>> Last update of whois database: 2022-09-16T23:47:24Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

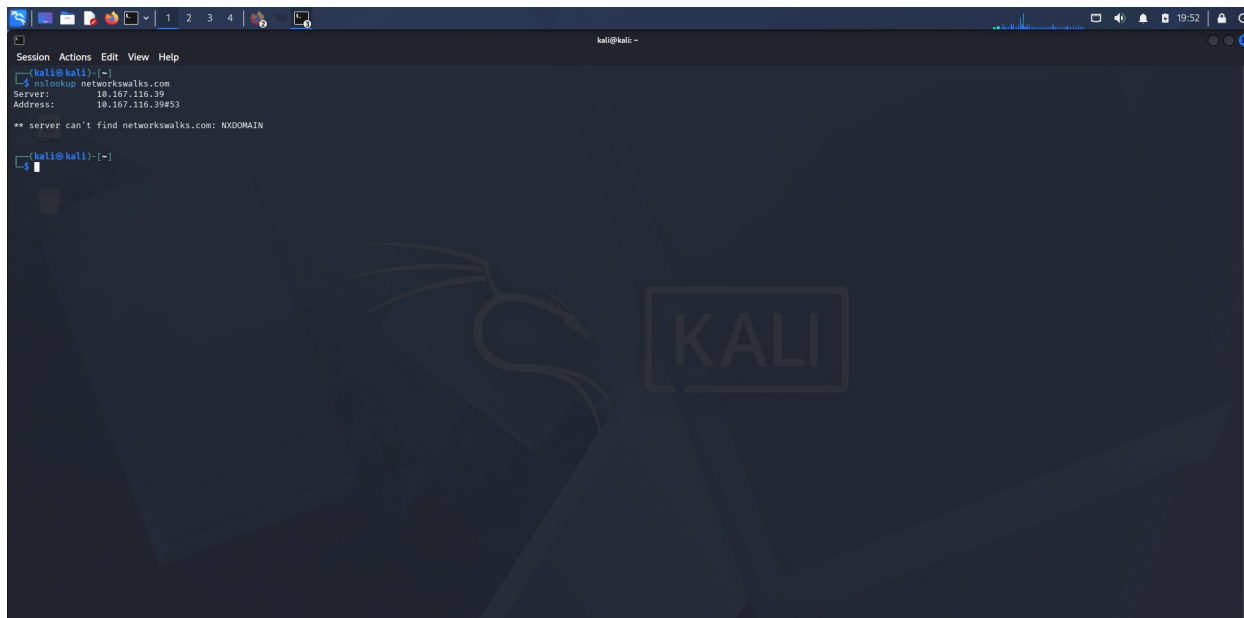
NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this data only
for lawful purposes and that under no circumstances will you use this data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
```

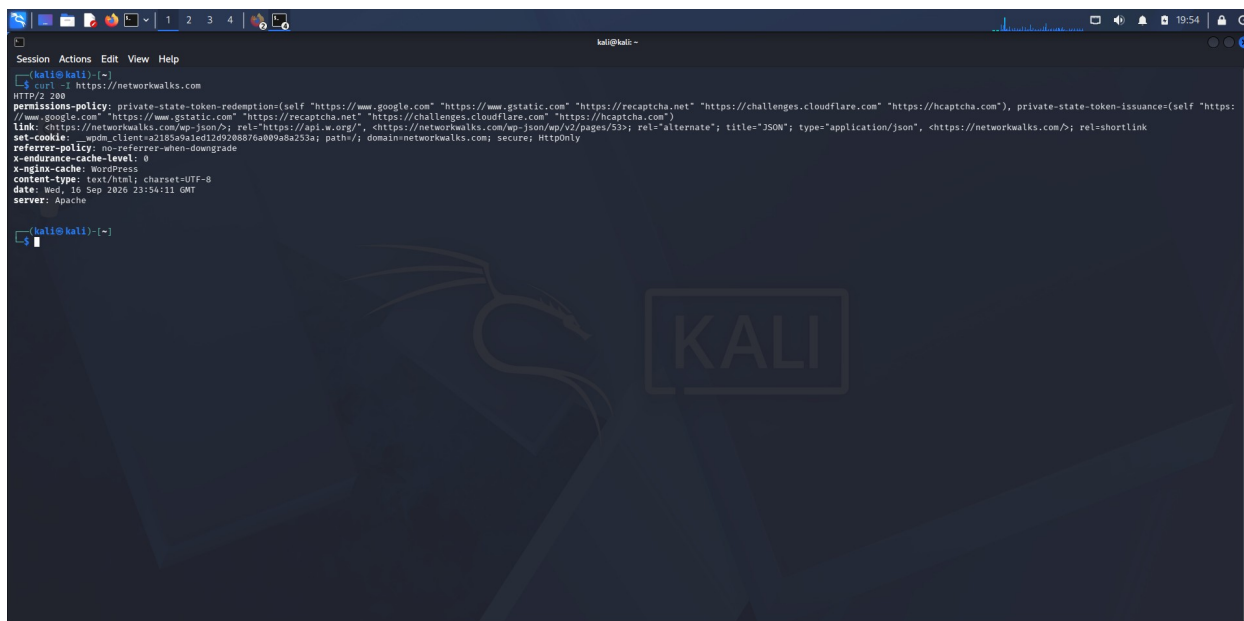
Evidence 2 — WhatWeb technology fingerprint (whatweb networkwalks.com)

```
kali@kali:~$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[wpdm_client], Country[UNITED STATES][🇺🇸], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by-upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[_wpdm_client], Country[UNITED STATES][🇺🇸], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1], WordPress Download Manager [3.3.5], Open-Graph-Protocol[website], Script[46849e-IPIDcompipidvar2=96510ampiprVar2=76ampscvvar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][🇺🇸], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1], WordPress Download Manager [3.3.5], Open-Graph-Protocol[website], Script[46849e-IPIDcompipidvar2=96510ampiprVar2=76ampscvvar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```

Evidence 3 — Nslookup query (nslookup networkwalks.com — NXDOMAIN due to typo)

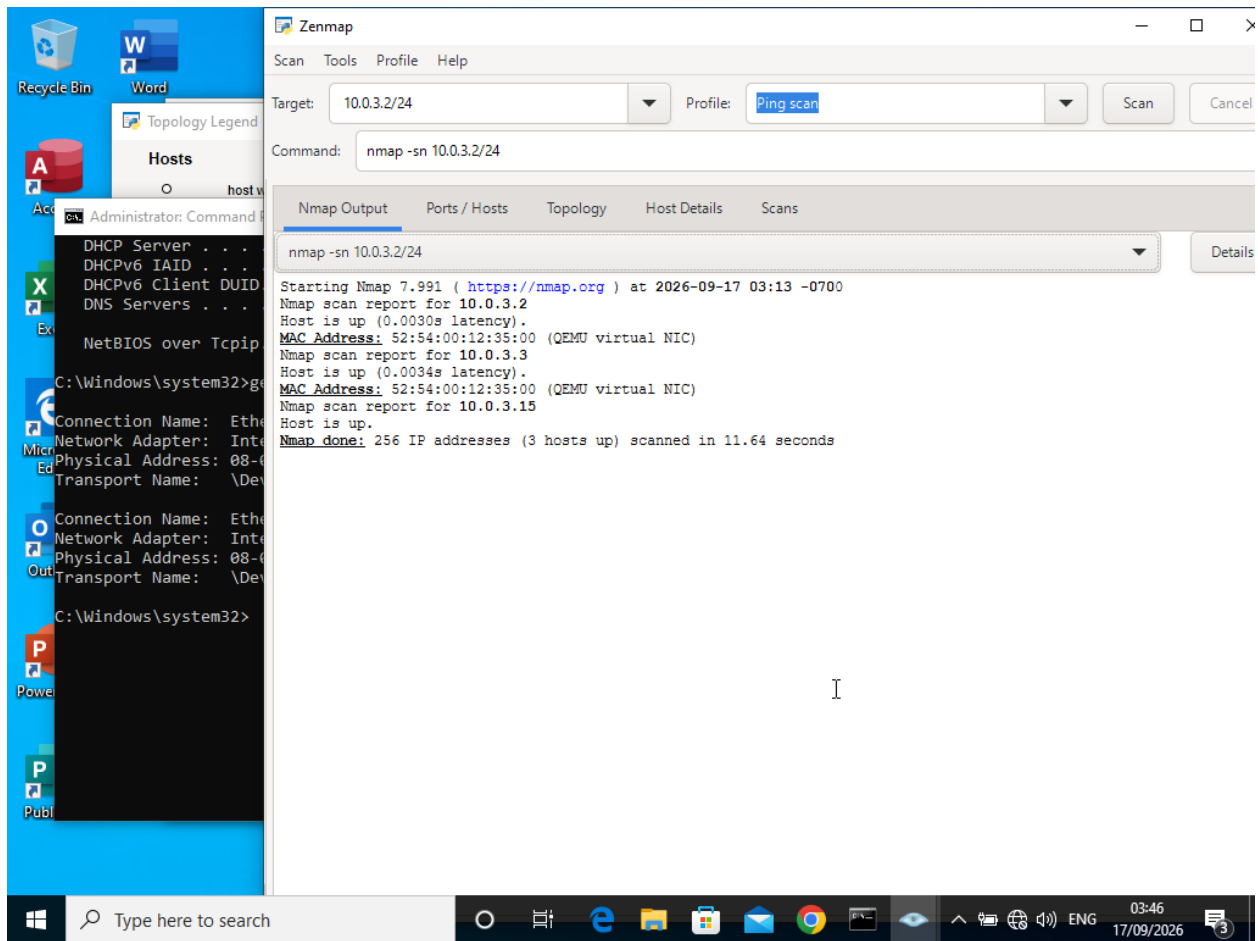


Evidence 4 — HTTP headers via Curl (curl -I https://networkwalks.com)

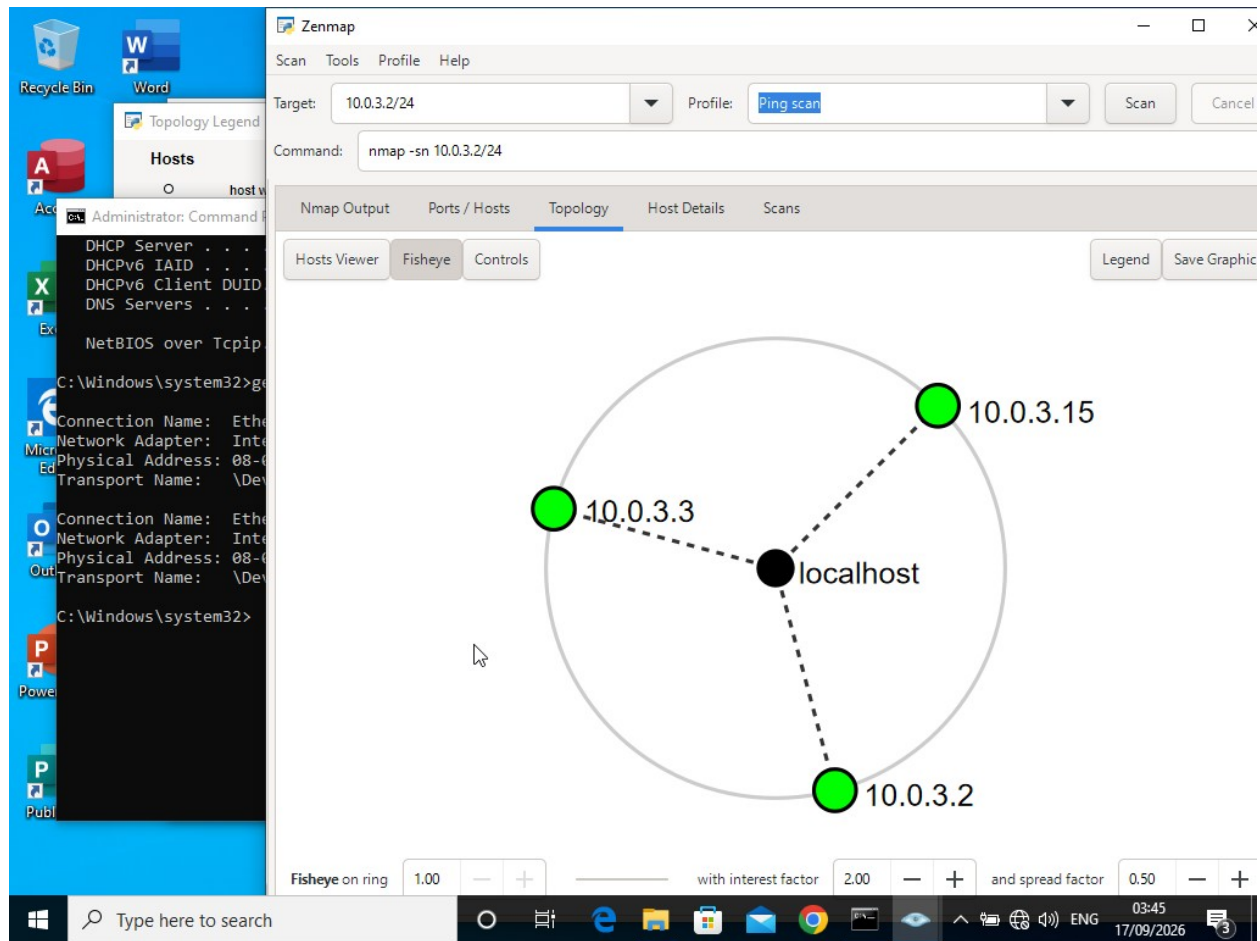


Evidence 5 — WAF detection via Wafw00f (wafw00f networkwalks.com)





Evidence 8 — Zenmap network topology (fisheye view)



-End-



Author

Sondasi

Cybersecurity Professional, Batch B083



Project Information

Program Name: Cybersecurity program at Networkwalks | Week: 02 | Repository: GitHub